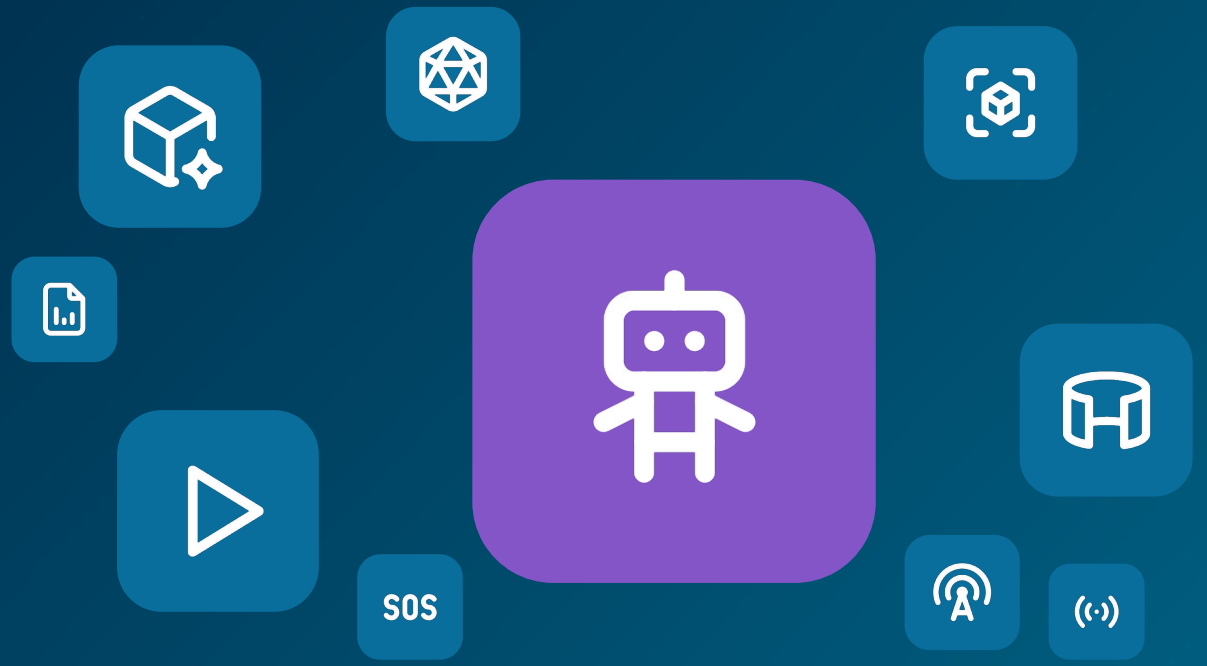
The Media Connectivity Association




OVERVIEW AND UPDATES · IMMERSIVE MEDIA

# Conversational Avatar Communication

Real-time conversational avatars, animated from user to user

MPEG (ISO/IEC) 3GPP

NOVEMBER 2026 – www.5g-mag.com

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# About the project

Real-time conversational avatars, animated from user to user

## 3GPP TR 26.813

**Technical Report**

Formats for representing and animating avatars, and recommendations for integrating them in 3GPP services. V20.0.0 is a Release 20 report; the studies are FS_AVATAR (Release 19) and FS_Avatar_Ph2_MED (Release 20, complete).

## MPEG ARF

**ISO/IEC 23090-39**

The Avatar Representation Format: base avatar data model, container and animation stream.

## 3GPP TS 26.264

**Technical Specification**

IMS-based AR real-time communication, created in Release 18; Release 19 added avatar calls over the IMS data channel and the three animation modes. Release 20 (work plan of 28 September 2026): AvCall_Ph2-MED, planned finish 3 March 2027.

FIND OUT MORE Technology Specifications Reference Tools

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# The avatar format: MPEG ARF

ISO/IEC 23090-39, as described in 3GPP TR 26.813

**Container** ISOBMFF or Zip

- ARF document
- Animation track (optional)

**Base Avatar Format** stores the avatar asset

- Skeletons
- Meshes
- Blendshapes
- Skins
- Landmarks
- Textures
- Nodes (joints)

**Animation Stream Format** live or stored

- Avatar Animation Units
- AAU header and payload
- configuration
- blendshape
- joint
- landmark
- texture

The base avatar and the animation are separate: a client loads the model once, then receives only the compact animation stream during the call.



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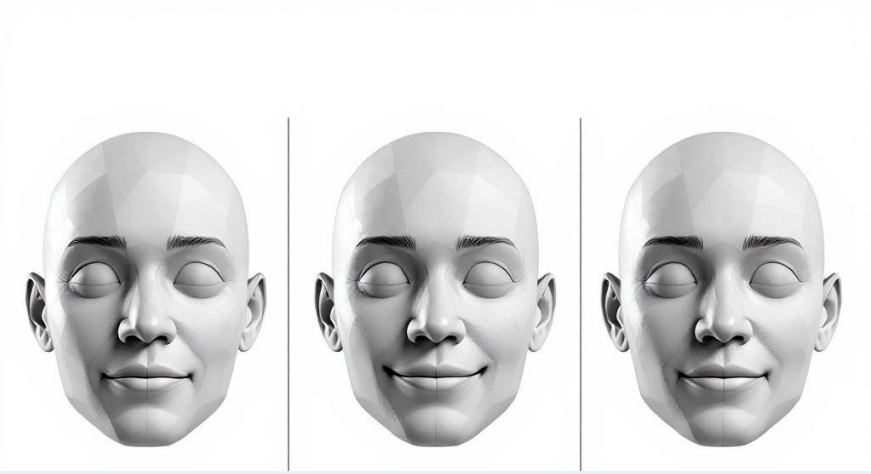
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# Animating the avatar

Avatar Animation Units (AAU) of 3GPP TR 26.813

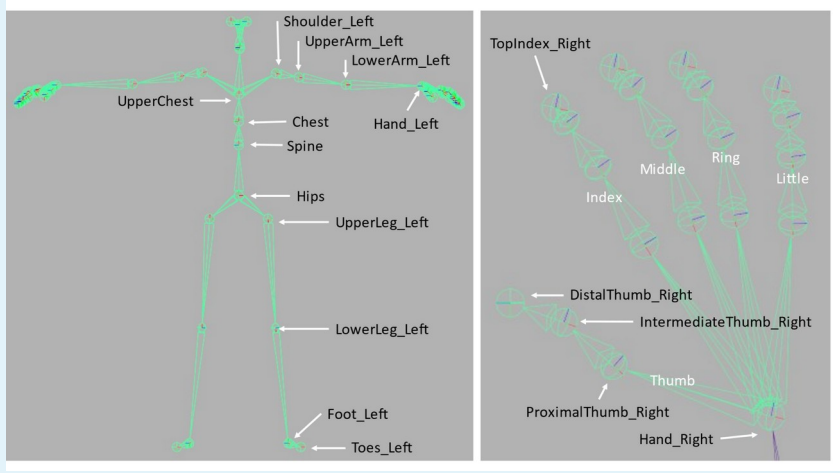
FIVE KINDS OF AAU

configuration | blendshape | joint | landmark | texture



## Facial animation

Blend shape animation starts from the neutral expression and adds weighted contributions from predefined expressions.




## Body and hand animation

Linear blend skinning moves the avatar surface with the skeleton, weighted per joint.

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# Three avatar animation modes

Three avatar animation modes defined by 3GPP TS 26.264 for a call over the IMS data channel

SENDER-CENTRIC

## The sender animates

"the sender UE animates and renders its avatar representation before sending it to the receiving UE as 2D video."

RECEIVER-CENTRIC

## The receiver animates

"the receiving UE animates and renders the sender UE's avatar representation."

5G-MAG Reference Tools

NETWORK-CENTRIC

## The network animates

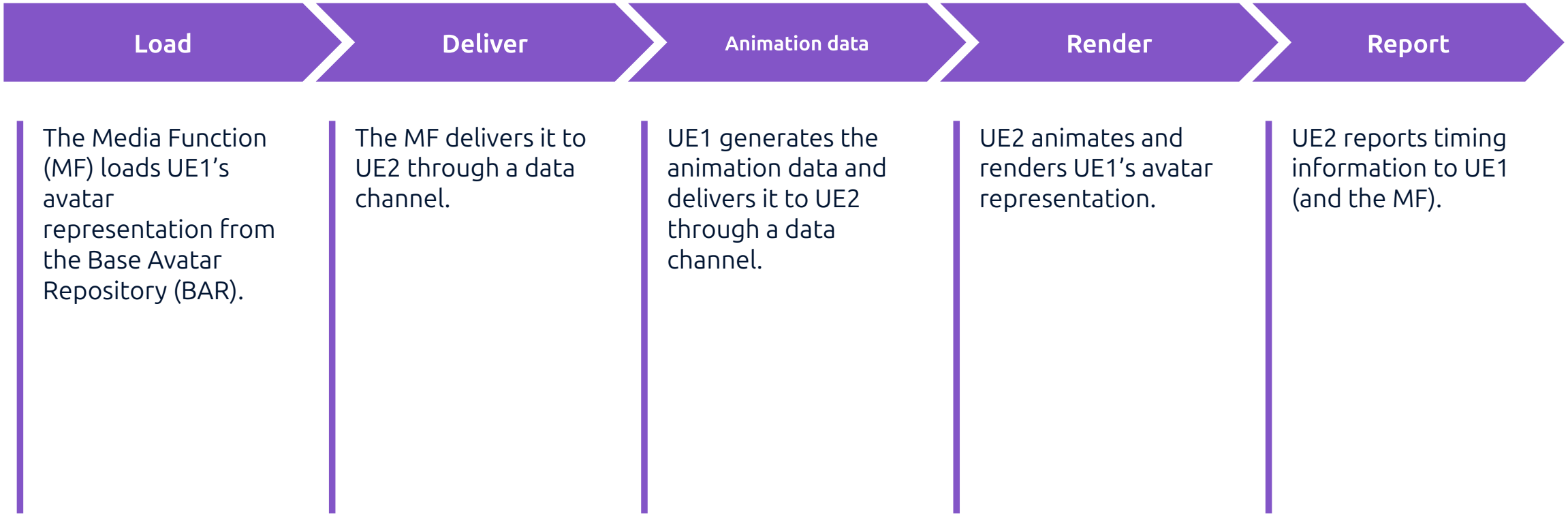
"the MF animates and renders the sender UE's avatar representation before sending it to the receiving UE as 2D video."

**Base Avatar Repository: "stores the base avatar of a user and makes it available to other participants" (3GPP TR 26.813).**

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# A receiver-centric call

Avatar delivery and animation flow of 3GPP TS 26.264, receiver-centric mode






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# Repositories

On main: a WebRTC call signalled through SWAP, carrying audio and avatar animation; IMS and network roles are not implemented

| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| rt-avatar-call | the Unity call application: Android and Windows | Receiver-centric call, ARF format, Animation, SWAP signalling | v0.10.154 16 Oct 2026 | GitHub |
| rt-avatar-content | reference MPEG ARF content for the call | ARF format | v0.10.154 16 Oct 2026 | GitHub |
| rt-avatar-unity-tracker | a sample OpenXR tracker for ARF avatar animation in Unity; no code yet | Animation | none yet | GitHub |
| rt-avatar-bar | stores base avatar models as ARF containers, scoped to a call; client fetch pending | Base Avatar Repository | none yet | GitHub |
| rt-3gpp-swap | the 3GPP SWAP server and Node.js library (TS 26.113) | SWAP signalling | v0.9 16 Oct 2026 | GitHub |

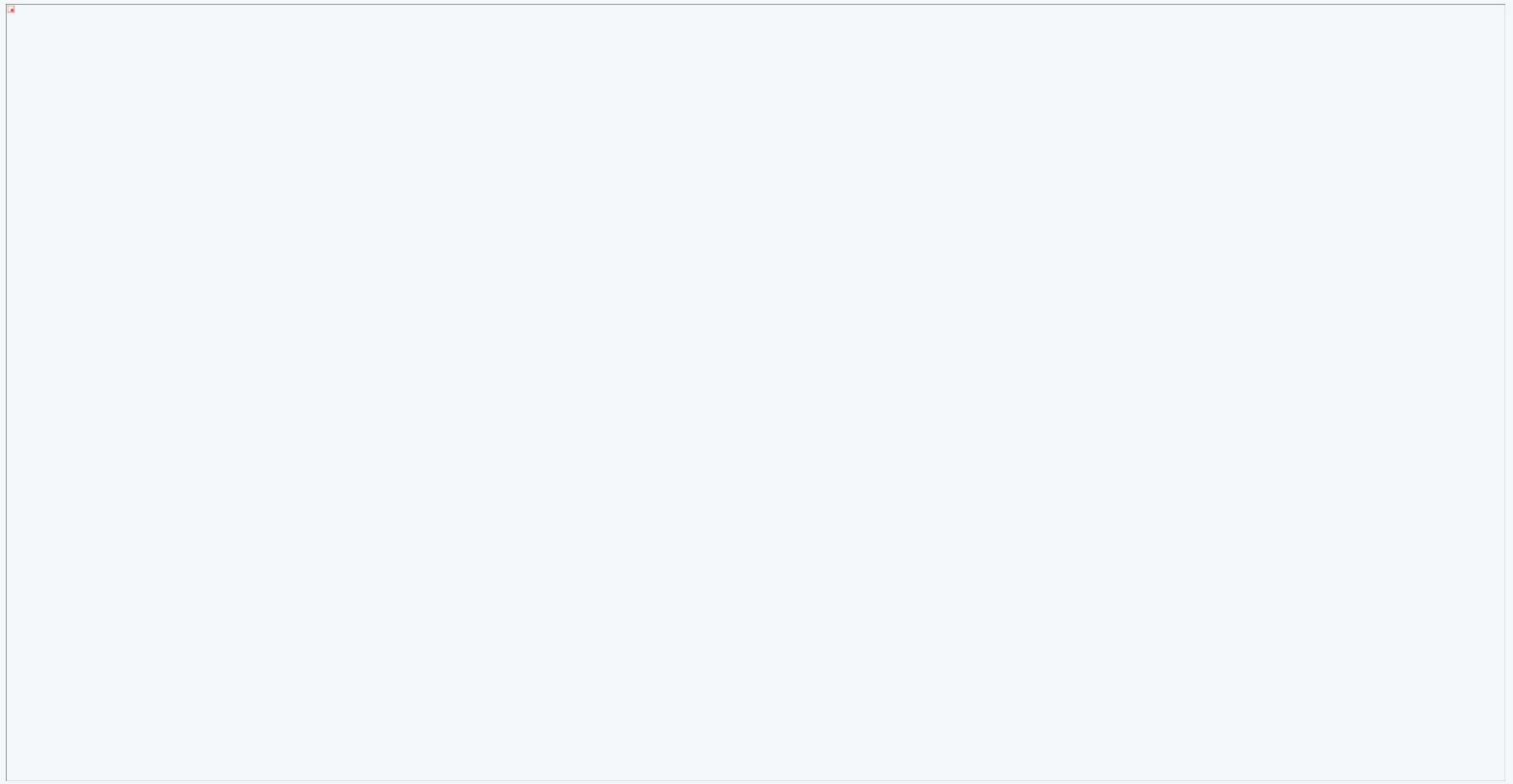
FIND OUT MORE **Technology** **Specifications** **Reference Tools**

● released ● no release yet

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# Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

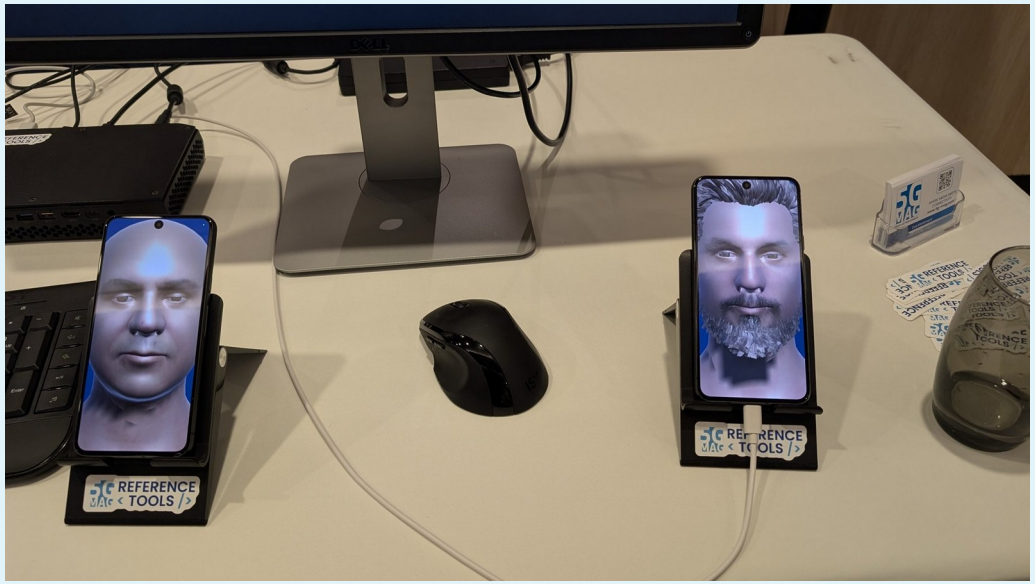
1. **rt-avatar-call**
2. **rt-avatar-content**
3. **MediaPipe**
   dependency, used by 1
4. **UnityARF**
   dependency, used by 1

Boxes without a badge are not mapped to a repository.

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# See it and try it

Photos of the project's demo, an avatar call on two phones




**MPEG meeting, Geneva, 14 July 2026**



**IBC 2026, 13 September 2026**

CONTRIBUTORS MOTION SPELL Qualcomm interdigital.

WHERE WE STAND

# Conversational Avatars with MPEG ARF

| BROUGHT TO STUDY | ON-GOING | OUTCOMES |
| --- | --- | --- |
| Avatar calls phase 2 | Gaussian splatting avatars | Initial Avatar Call Client |
| Body animation support in the avatar call client | Animation generation on Android XR | Initial implementation of SWAP protocol |
| | Base Avatar Repository | Avatar content for the call client |
| | Android Avatar Call Client Application with enhanced UX | |

RELATION WITH SDOS AND INDUSTRY FORA MPEG (ISO/IEC) 3GPP

**MPEG** Avatar part of contribution m78825 at MPEG#156 (Hangzhou, October 2026): a receiver-centric avatar call on two phones

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